PYTHON OUTPUT / CASE STUDY 1 OF 10

```
======================================================================
23. STUDENT TIMETABLE CONFLICT DETECTION
======================================================================
Course timetable:
  Python   Mon periods 1-2, room R101
  Maths    Mon periods 2-3, room R102
  English  Mon periods 4-5, room R101
  Physics  Tue periods 1-2, room R103
CONFLICT: Python overlaps with Maths on Mon.
Unavailable-room assignments: None
Timetable status: INVALID

======================================================================
```

PYTHON OUTPUT / CASE STUDY 2 OF 10

```
======================================================================
28. REPEATED TRANSACTION DETECTION
======================================================================
Transaction records:
  TX101: Rs. 450
  TX102: Rs. 1200
  TX101: Rs. 450
  TX103: Rs. 9500
  TX104: Rs. -50
  TX105: Rs. 700
Suspicious entries:
  TX101 | Rs. 450 | duplicate ID
  TX103 | Rs. 9500 | above threshold
  TX104 | Rs. -50 | negative amount
Total records: 6
Suspicious records: 3

======================================================================
```

PYTHON OUTPUT / CASE STUDY 3 OF 10

```
======================================================================
30. STUDENT EMAIL AND IDENTIFIER VALIDATION
======================================================================
Email checks:
  rehan23@college.edu        -> VALID (valid format)
  aisha.k@college.edu        -> VALID (valid format)
  bad-email                  -> INVALID (must contain exactly one @)
  student@@college.edu       -> INVALID (must contain exactly one @)
  name@college               -> INVALID (domain must contain a dot in a valid position)
Registration ID checks:
  CSE2026001     -> VALID (valid format)
  ECE2026042     -> VALID (valid format)
  CS2026         -> INVALID (must be 10 characters)
  CSE20A6003     -> INVALID (last 7 characters must be digits)

======================================================================
```

```
======================================================================
38. ATM MENU-DRIVEN BANKING SIMULATION
======================================================================
Deposit Rs. 1500.00: accepted
Withdrawal Rs. 1200.00: accepted
Withdrawal Rs. 2500.00: rejected (daily limit exceeded)
Balance inquiry: Rs. 5300.00
Transaction summary:
  Deposit    Rs.  1500.00 - Accepted
  Withdrawal Rs.  1200.00 - Accepted
  Withdrawal Rs.  2500.00 - Rejected
Final balance: Rs. 5300.00
Total withdrawn today: Rs. 1200.00

======================================================================
```

PYTHON OUTPUT / CASE STUDY 5 OF 10

```
======================================================================
66. STUDENT ACADEMIC RISK CLASSIFICATION
======================================================================
Risk rules: marks average < 40, attendance < 75%, lab score < 50%.
Asha     average= 77.0, attendance= 92%, lab= 85%
         risk=LOW; factors=none
Bharat   average= 43.0, attendance= 68%, lab= 45%
         risk=HIGH; factors=low attendance, low laboratory score
Charan   average= 58.0, attendance= 76%, lab= 52%
         risk=LOW; factors=none
Divya    average= 39.7, attendance= 82%, lab= 38%
         risk=HIGH; factors=low marks, low laboratory score
Summary: HIGH=2, MEDIUM=0, LOW=2

======================================================================
```

PYTHON OUTPUT / CASE STUDY 6 OF 10

```
======================================================================
72. SMART HOSTEL RESOURCE MONITORING
======================================================================
Room 101 (floor 1): 145 kWh, 1800 L, bill=Rs. 1196.00; alerts=none
Room 102 (floor 1): 95 kWh, 1200 L, bill=Rs. 784.00; alerts=none
Room 201 (floor 2): 210 kWh, 2400 L, bill=Rs. 1728.00; alerts=electricity high, water
    high
Room 202 (floor 2): 130 kWh, 1600 L, bill=Rs. 1072.00; alerts=none
Floor summary:
  Floor 1: 240 kWh, 3000 L, total charges Rs. 1980.00
  Floor 2: 340 kWh, 4000 L, total charges Rs. 2800.00

======================================================================
```

PYTHON OUTPUT / CASE STUDY 7 OF 10

```
======================================================================
76. UNIVERSITY EXAMINATION RESULT MODERATION
======================================================================
Asha: original=[38, 67, 81], moderated=[41, 67, 81]
  Result before moderation: FAIL
  Result after moderation:  PASS
Bharat: original=[29, 45, 54], moderated=[32, 45, 54]
  Result before moderation: FAIL
  Result after moderation:  FAIL
Charan: original=[49, 52, 61], moderated=[49, 52, 61]
  Result before moderation: PASS
  Result after moderation:  PASS
Rule: 3 grace marks are added only to subjects below 40; maximum is 100.

======================================================================
```

PYTHON OUTPUT / CASE STUDY 8 OF 10

```
======================================================================
77. SMART ELECTRICITY BILLING FOR MULTIPLE APARTMENTS
======================================================================
A-101: 85 units; energy=Rs. 255.00; fixed=Rs. 50; rebate=Rs. 0; bill=Rs. 305.00
A-102: 145 units; energy=Rs. 525.00; fixed=Rs. 50; rebate=Rs. 0; bill=Rs. 575.00
B-201: 230 units; energy=Rs. 1010.00; fixed=Rs. 50; rebate=Rs. 0; bill=Rs. 1060.00
B-202: 55 units; energy=Rs. 165.00; fixed=Rs. 50; rebate=Rs. 20; bill=Rs. 195.00
Apartments ranked by consumption (highest first):
  B-201: 230 units
  A-102: 145 units
  A-101: 85 units
  B-202: 55 units

======================================================================
```

PYTHON OUTPUT / CASE STUDY 9 OF 10

```
======================================================================
81. CAMPUS EVENT SEATING AND ENTRY VALIDATION
======================================================================
Seat map:
  Row 1: REG101 | REG102 | EMPTY
  Row 2: REG103 | REG104 | REG105
  Row 3: EMPTY | REG106 | REG102
Ticket validation:
  REG101: ACCEPTED
  REG102: ACCEPTED
  REG103: ACCEPTED
  REG102: REJECTED (duplicate entry)
  REG999: REJECTED (registration not seated)
Total capacity: 9
Occupied seats: 7
Available seats: 2
Duplicate seat assignments: ['REG102']

======================================================================
```

PYTHON OUTPUT / CASE STUDY 10 OF 10

```
======================================================================
94. DISASTER RELIEF SUPPLY ALLOCATION
======================================================================
Allocation policy: priority 1 is most urgent; allocate available stock in priority
    order.
Area A  | Water      | priority 1 | needed  70 | allocated  70 | unmet   0
Area A  | Food packs | priority 1 | needed  50 | allocated  50 | unmet   0
Area B  | Blankets   | priority 1 | needed  30 | allocated  30 | unmet   0
Area B  | Water      | priority 2 | needed  50 | allocated  30 | unmet  20
Area C  | Food packs | priority 2 | needed  45 | allocated  30 | unmet  15
Area C  | Blankets   | priority 3 | needed  20 | allocated  10 | unmet  10
Remaining stock:
  Water: 0
  Food packs: 0
  Blankets: 0
Total unmet demand: 45

All 10 case-study demonstrations completed.
```